

Paths Through Spatial Networks

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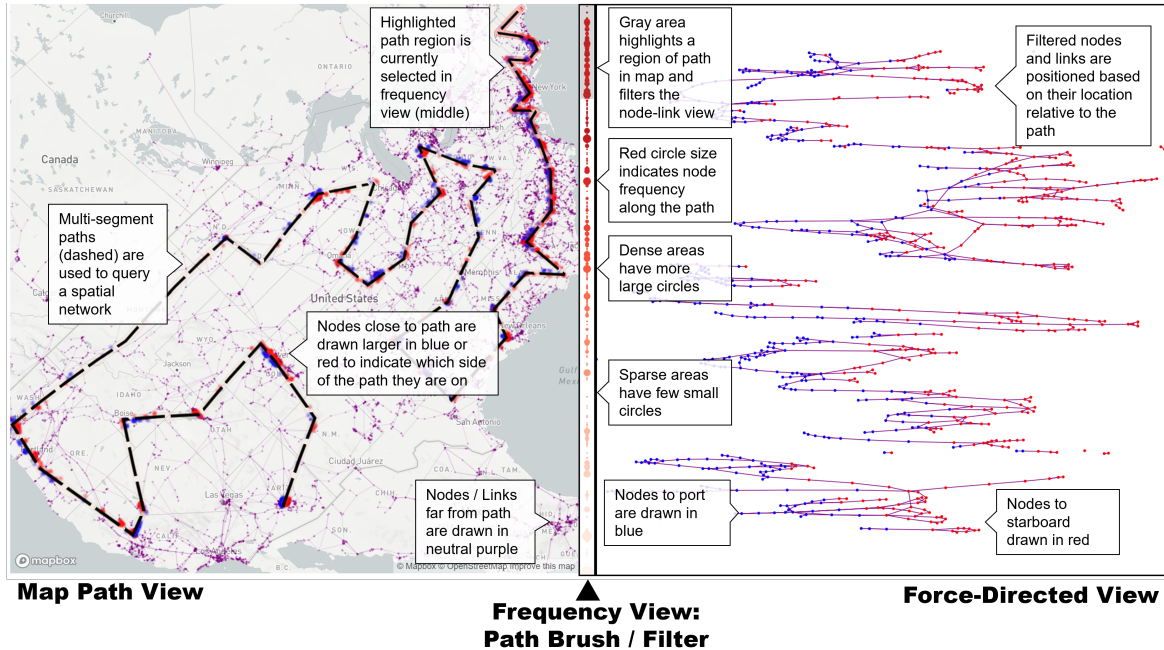


Figure 1: The power grid of North America explored using a path connecting the lower 48 states of the U.S. On the left, a map visualization of the path and the nearby power grid nodes is displayed as an overview. In the middle, a representation of the node frequency along the path is used to brush and filter for the subsection of the path from Atlanta, GA to Augusta, ME. On the right, a detailed force-directed node-link diagram is given for the selected portion of the path, indicating disconnected network components.

ABSTRACT

Spatial networks present unique challenges to understanding topological structure. Each node occupies a location in physical space (e.g., longitude and latitude) and each link may either indicate a fixed and well-described path (e.g., along streets) or a logical connection only. Placing these elements in a map maintains these physical relationships while making it more difficult to identify topological features of interest such as clusters, cliques, and paths. While some systems provide coordinated representations of a spatial network, it is almost universally assumed that the network itself is the sole mechanism for movement across space. In this paper, I present a novel approach for exploring spatial networks by orienting them along a point or path in physical space that provides the guide for parameters in a force-directed layout. By specifying a path across topography, networks can be spatially filtered independently of the topology of the network. Initial case studies indicate promising results for exploring spatial networks in transportation and energy distribution.

Index Terms: Human-centered computing—Visualization—Visualization techniques—Graph Drawing; Human-centered

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1 INTRODUCTION

Decisions about the network structure, whether operating on it or moving through it, need to be made with respect to not only the importance of the nodes and links to the overall network structure (e.g., the centrality of a node, bottlenecks on links with high betweenness), but the location of the network components in the real world. Part of the challenge of examining a spatial network in a network layout (e.g., force-directed [8]) is that the given locations of nodes and links are meaningful in the map. The spatial characteristics of the network are heavily influenced by nearby locations [2], and provide meaningful insight into the overall structure of the network. As Tobler remarked, “everything is related to everything else, but near things are more related than distant things.” [15]. Nearness has two meanings in a spatial network: distance in the real world, and distance along the network. While many network layout techniques serve well to explore relationships in the latter, they do so at the expense of the former. Consequently, allowing a non-deterministic technique to generate a good layout of the given nodes may be useful in understanding the topology, but the resulting node placements lose all meaning with respect to their original placement on the map due to the distortion of relationships of location. This makes it nearly impossible to easily translate the insight gained from the network layout back to the original spatial context. While these aspects are critical, their spatial characteristics must not be sacrificed.

To address these challenges, I present a view composition technique to analyze spatial networks. Paths through the spatial network are specified by the user to construct a placement axis within a force-directed layout. These paths can be constructed from a single point, a straight line, or more complex lines involving turns. The relative distance and orientation of the nodes with respect to the drawn path are used to generate a force-directed layout that yields topological insight while maintaining a correlation to the original spatial context. This technique has been implemented in an initial prototype in D3 [3].

2 RELATED WORK

There is no shortage of existing visualization techniques for geospatial networks. In their survey of the domain, Schöttler et al. identify no fewer than ninety-five techniques for exploring this design space, highlighting the tension and trade-offs between maintaining geospatial fidelity and emphasizing network structure [12]. Relating this to the work of Javed and Elmqvist exploring composite visualization, they point out the benefits of coordinated views in that the strong location fidelity of a geospatial representation can be coupled with a more abstract representation of the network [10]. Network data adds the additional wrinkle in that much of the information of interest is inherently non-spatial, requiring understanding of clustering and connectivity relationships between groups of disparate nodes. For example, Zeng et al. utilized origin-destination overlays on a flow map tree to indicate the relative efficiency of potential paths in a public transportation network across accessible regions [19].

While these abstract representations provide benefits in the context of the tasks they support, it remains critical to maintain a connection to the original spatial structures [1]. In systems like MapTrix, the coupling relies only on link marks between related regions and requires no interaction [18]. Alternatively, systems like Flowstrates allow the user to specify a single origin and destination and examine attributes over time for the routes between those two locations [4]. While this range of options provides a meaningful exploration of the design space, what is less commonly addressed is how to explore networks in a way that is not constrained to the network being examined. For travel networks involving multiple connected locations, Shifted Maps provides interactive flexibility in rearranging the spatial locations into a more user-oriented representation of the topology [11], and moves towards away from constraining the exploration of the data strictly within predefined travel networks. Similarly, the proposed technique removes the constraint of moving through the network, instead allowing the user to define their own path and filtering the network independently of the topological structure of the network.

Prior research has explored user-specified paths and shapes to analyze other types of spatial data. Brosz et al. suggested the use of transmorphifiers to map arbitrary spatial regions onto new layouts through raster deformation, such as mapping a curved line across a map into an axis-aligned rectangle [5]. The correspondence between the input representation and the output layout maintains the spatial context. Turkay et. al [16] utilized a sketched path to analyze multivariate change in data values and distributions over the course of the path as attribute signatures. In this case, the user is able to understand potential relationships between related attributes while moving across a map. In HotSketch [9], a user draws a path to aggregate crime data near the path and understand the relative distribution of crime events and types along it, observing the relative change in frequency between types of crimes that occur during daytime and those that occur at night. StreetVizor utilized a path through the city to aggregate data and images that are presented contextually to a user in a coordinated view [13]. In these cases, and in many similar systems, the aggregated data is tabular. While some systems provide coordinated representations of a spatial network, it is almost universally assumed that the network itself is the sole

mechanism for movement across space.

3 DERIVING THE PLACEMENT AXIS AND FREQUENCY VIEW

The proposed technique relies on a placement axis to provide force-directed guidance to a non-deterministic layout algorithm, transforming the straight or curving paths across a map into a linearized form with a consistent x and y axis. These paths can be constructed from a single point, a straight line, or more complex lines involving turns. In all cases, the technique requires a radius to filter the points in the network, only incorporating those points that are within that distance from the path (i.e., a buffer or query region - see Figure 2). This buffer is rendered with a translucent yellow fill on the map. In the case of a point, the system extends a search line at bearing 0° (due North) and bearing 180° (due South) in order to construct a rough path at the required distance (Figure 3). For simple and complex lines, the beginning and end of the path are similarly extended. Then, paths are segmented into evenly spaced intervals, which facilitates the construction of a sequence of waypoints along the path that can be used to position the nodes in the y-axis.

3.1 Y-Axis: Clustering Points along a Path

Given a path, each node within the buffer is compared to the path to determine the point along the path that is closest to the node and construct the frequency view. While in lower-density areas these nodes along the path can be quite spaced out, in some cases, particularly the endpoints, they can be overplotted and heavily redundant as individual nodes might be close to many different segments. To efficiently and uniquely attribute nodes to segments and evaluate changes in frequency across the path length, the nodes and segments are clustered using DBSCAN [7]. Given the closest segment along a path for a node, nodes can be positioned correctly along the y-axis based upon their distance from the starting point: nodes encountered sooner along the path are positioned at the bottom of the axis and nodes closer to the end are at the top.

The y-axis allows the path to be linearized in the frequency view: circles representing the waypoints can be positioned vertically and used to filter the force-directed view (middle column of Figure 1). The radius of each representative circle is scaled to indicate the number of nodes clustered at that waypoint, and the circles themselves are shaded using a monotonically increasing red color scale from the bottom to top for separation and to reinforce the overall forward trajectory of the path. In this interface, a boxed region can be dragged and resized to brush over the path, following the mantra "overview first, zoom and filter, details on demand." [14]. As the user filters the frequency view, a secondary red buffer layer is drawn on top of the path and selected nodes are highlighted in the map view to indicate the portion that is currently being filtered spatially. Similarly, only the points within the brushed region are used to populate the force-directed layout.

3.2 X-Axis: Node Orientation Relative to Path

To finish the force-directed view, we utilize the distance and orientation between a node and its closest path segment to determine the magnitude and direction of its x-axis position. To determine its direction, we must obtain the orientation of the node with respect to the path through space. In nautical terms, if the path is a ship travelling through the ocean, is the node to port (left side facing towards the front of the boat) or to starboard (right)? To determine this orientation, we identify which segment of the path closest to the node. We then construct a polygon consisting of the starting coordinates for that segment, the node, and the endpoint to determine if the coordinates are clockwise. If the points are clockwise, the node is to "port" and if counter-clockwise the node is to "starboard." To help reinforce this orientation and demonstrate the preservation of spatial context throughout the figures in this paper, port nodes are rendered in blue and starboard nodes are rendered in red (Figure 2).

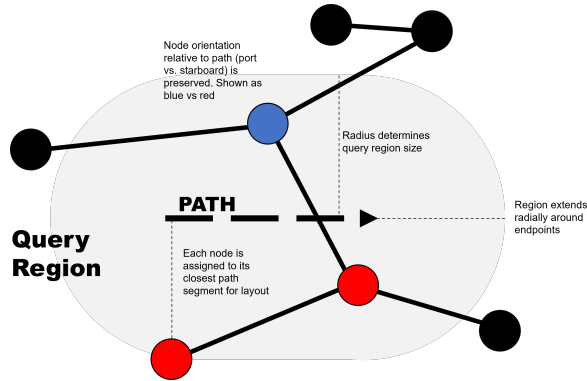


Figure 2: Each node within the query region is assessed to determine the point along the path that it is closest to and the distance to that point. The region encloses the path and its endpoints in a radial envelope. Points associated with the same endpoint are clustered to minimize redundancy and reveal frequency in the brush and filter view.

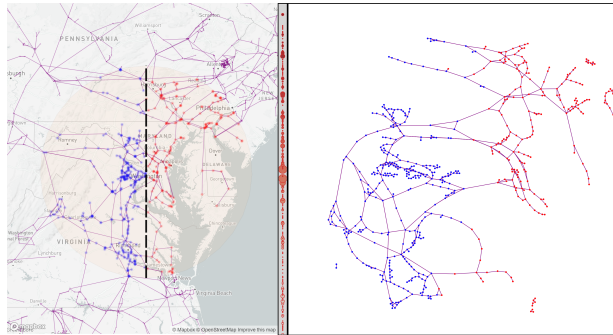


Figure 3: The power network around Washington, D.C. is explored at a 200km radius. Given a single point, the system extends a search line at bearing 0° (due North) and bearing 180° (due South). The resulting path filters the points within the search radius of the point (yellow circle) and defines a placement axis for the frequency and force-directed views. This reveals two major energy components around the DC area: one to the Northeast and one to the Southwest.

4 FORCE-DIRECTED LAYOUT

Given the position of the waypoints along the length of the path, distance from the line, and relative orientation, we have an initial framing for providing force-directed alignment to the nodes within the brushed region along the path. These parameters provide rendering hints for a 2D representation, in which distance to the node's closest segment from the start encodes a suggested y-position, distance from the path encodes an x-position, and port/starboard orientation encodes the signed distance (+/-) of an x-position relative to center. Consequently, in our force simulation, each node is operated on in concert by a collection of individual forces:

- A charge force that pushes nodes apart from each other.
- A link force that ensures that nodes that are connected to each other are pulled together.
- A y-axis force that pushes the node towards a scaled vertical position as a function of the ordinal index of its waypoint in the list of currently brushed waypoints.

- An x-axis force that pushes the node towards a scaled horizontal position from the midpoint of the network layout. The magnitude of the force is the node's distance from the path and the direction is determined by the port / starboard side orientation from the path.
- A collision force that pushes nodes away to avoid overplotting.

Taken together, these forces provide both x and y axes for orienting the position of the nodes in positions that closely resemble their spatial arrangement without necessarily forcing them to exactly mimic it (Figure 3). As previously mentioned, the coloring of the nodes in these figures is used to indicate the relative orientation of the node with respect to the path. In many cases, however, a single red node can be found quite close to the position of a cluster of blue nodes. In circumstances like this, the link forces of the blue nodes have overwhelmed the x-axis force of the single red node, pulling it over towards the rest of the nodes that are nearby topologically, if not spatially. While a compromise of the spatial rendering for the topological benefits, it is not a complete abandonment of the spatial representation, as it is still relatively easy to examine the side-by-side views of the coordinated views. To enhance coordination within the views, nodes are added to the top of the force-directed view when the filter box in the frequency view is dragged or expanded upward and added to the bottom when dragged or expanded downward. Nodes and their links are dynamically added and removed as the user brushes, and the highlighted portion moves across the path in the map view as the user brushes.

5 CASE STUDY - POWER GRID

To explore the validity of the proposed technique, we will now explore the interface using a case study examining a spatial network that a person cannot move through - a continent-wide power network. While the technique can utilize more complex paths (Figure 1), the case below will utilize straight line paths only to ease understanding.

A user is looking for bottlenecks in the power grid network across a large swathe of the United States. Utilizing the Gridkit Energy Grid Dataset of North America [17] for analysis, they construct a path across the country from Washington, D.C. to Los Angeles, CA. The user is looking for locations in the energy grid that are vulnerable, either to ransomware attacks or to more natural disasters. These nodes will typically be present in isolated areas where there are relatively few connections to other major grid components, so a disturbance in the network could destabilize the local network's load shedding ability or cause brownouts during periods of peak demand.

The user examines the network by dragging the brushing control window to cover a less-populated region of the country in which there are fewer nodes but where they are tightly packed (Figure 4). The beginning of the selected area is in Missouri and the more Western endpoint ends at the Nevada-California border. While the network robust connectivity through Missouri, as the path proceeds through Kansas and into Colorado the network becomes more sparsely connected and relies on relatively few connections (red component). Along this 200km-wide path, the network tenuously connects through the southern edge of Colorado Springs to Las Vegas through very few possible links. Along this path, there are many communities which could be heavily disrupted by adverse events in the power grid, and should be scheduled for infrastructure review and improvement soon.

6 DISCUSSION AND FUTURE WORK

In this paper, I have proposed a straightforward and useful technique for constructing a path-aligned force-directed layout of a spatial network. While the initial use cases are promising, they utilize spatial networks in which most links are relatively short and represent the physical path that is taken by something to get from one node to another (i.e., a power line, a road). In other experiments, I have

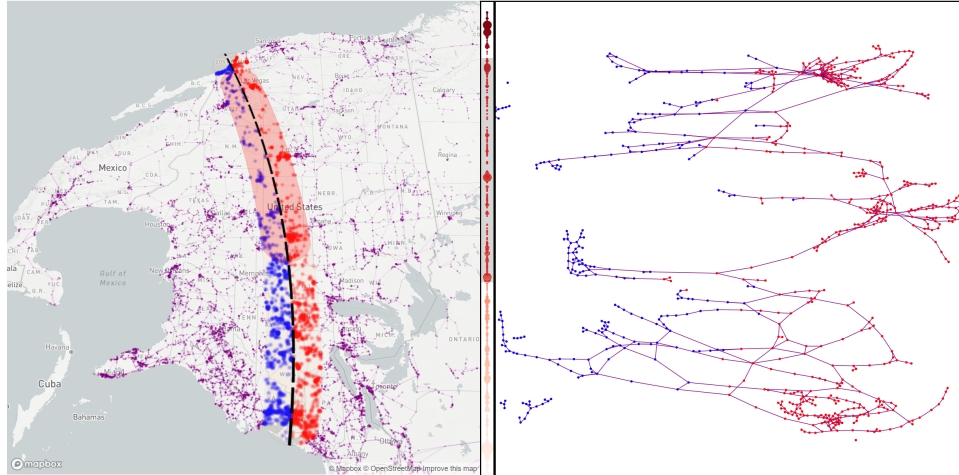


Figure 4: The route from Washington, DC to Los Angeles in the context of a power grid network. The user has selected a subset of the route in the relatively less-populated Midwestern region of the trip. The node-link diagram reveals three major clusters along this section of the route that are only tenuously connected by a single path.

explored the use of the proposed technique for a social network in which the physical locations of the nodes were spatial (specifically the Gowalla dataset from SNAP [6]), but the links were not restricted by any physical requirements - ranging over vast distances. In these cases, where the links of the nodes are more likely to cross from distant waypoints, the initial results are less promising.

Still, in the provided examples and similar cases, it seems like an initially promising step into mitigating the issues of consolidating spatial and non-spatial characteristics of networks. For future work, it would be useful to augment the force-directed layout with rendering cues to indicate to the user which nodes, and to what extent, have connections outside the currently filtered buffer region. This would help the user to better understand if some characteristics of the network are truly a property of the network topology or only of the currently defined filtering distance.

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